

$\tau(p; q)$ -summing and quasi- $\tau(p; q)$ -summing operators coincide isometrically

Candidate full-solution packet

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Status: candidate full proof likely valid; human review requested. The open problem in arXiv:1608.06926 has an affirmative isometric answer. More strongly, the equality holds for all mixed exponents $1 \leq q \leq p < \infty$ and in every multilinear degree.

1 Source question and result

Botelho and Mujica introduced quasi- $\tau(p)$ -summing operators in their duality theory for $\sigma(p)$ -nuclear operators [1]. At the end of the paper they asked:

“Is it true that $\mathcal{L}_{\tau(p)}(E; F') = \mathcal{L}_{q\tau(p)}(E; F')$ for all Banach spaces E and F ?”

The question appears on page 15 of the arXiv version (page 236 of the published version):

are each other's inverse in the sense that $\varphi_{S\varphi} = \varphi$ and $\psi\varphi_S = \psi$ for $\varphi \in [\mathcal{L}_{\sigma(p)}(E_1, \dots, E_n; F)]'$ and $S \in \mathcal{L}_{q\tau(p)}(E'_1, \dots, E'_n; F')$. The equality $\|S\varphi\|_{q\tau(p)} = \|\varphi\|$ completes the proof. $\square$

For reflexive target spaces we have:

3.5. Corollary. *If $E'_1, \dots, E'_n$ have the bounded approximation property and F is reflexive, then the spaces $[\mathcal{L}_{\sigma(p)}(E_1, \dots, E_n; F)]'$ and $\mathcal{L}_{\tau(p)}(E'_1, \dots, E'_n; F')$ are isometrically isomorphic via the Borel transform. In particular, if E' has the bounded approximation property and F is reflexive, then $[\mathcal{L}_{\sigma(p)}(E; F)]' \stackrel{1}{=} \mathcal{L}_{\tau(p)}(E'; F')$.*

Open problem. Is it true that $\mathcal{L}_{\tau(p)}(E; F') = \mathcal{L}_{q\tau(p)}(E; F')$ for all Banach spaces E and F ?

We prove an affirmative answer with equality of the defining norms. The argument also answers the corresponding mixed-exponent and multilinear question, which sharpens the statement asked in the source.

2 Definitions

Let $E_1, \dots, E_n, F$ be real or complex Banach spaces and $1 \leq q \leq p < \infty$. For

$$S \in \mathcal{L}(E_1, \dots, E_n; F')$$

the quasi- $\tau(p; q)$ inequality is

$$\begin{aligned} & \left(\sum_{j=1}^m |S(x_{1j}, \dots, x_{nj})(y_j)|^p \right)^{1/p} \\ & \leq C \sup_{\substack{x'_i \in B_{E'_i} \\ y' \in B_{F'}}} \left(\sum_{j=1}^m |x'_1(x_{1j}) \cdots x'_n(x_{nj})y'(y_j)|^q \right)^{1/q}, \end{aligned} \quad (1)$$

for all finite families $x_{ij} \in E_i$ and $y_j \in F$. The least C is $\|S\|_{q\tau(p;q)}$ [1, Definition 3.1].

Mujica's ordinary $\tau(p; q)$ definition [2, Definition 3.2], specialized to a map with target F' , requires instead

$$\begin{aligned} & \left(\sum_{j=1}^m |y''_j(S(x_{1j}, \dots, x_{nj}))|^p \right)^{1/p} \\ & \leq C \sup_{\substack{x'_i \in B_{E'_i} \\ y' \in B_{F'}}} \left(\sum_{j=1}^m |x'_1(x_{1j}) \cdots x'_n(x_{nj})y''_j(y')|^q \right)^{1/q}, \end{aligned} \quad (2)$$

for arbitrary $y''_j \in F''$. The least C is $\|S\|_{\tau(p;q)}$. Thus the quasi definition is exactly the restriction of the ordinary definition to $y''_j = J_F y_j$, where $J_F : F \rightarrow F''$ is the canonical embedding. In particular,

$$\|S\|_{q\tau(p;q)} \leq \|S\|_{\tau(p;q)}. \quad (3)$$

3 Proof intuition

Only a finite family of bidual test vectors occurs in any instance of (2). Local reflexivity transfers their finite-dimensional span into F with arbitrarily small norm loss, while preserving exactly the pairings with the finitely many values of S . The ordinary numerator then becomes a quasi numerator without error.

The denominator is subtler. Composing a functional in F' with the local reflexivity map initially gives a functional only on the chosen subspace of F'' . Hahn–Banach extends it to F''' , and Goldstine density approximates that extension, on the finite test family, by canonical elements of F' . This turns the operator norm bound from local reflexivity into the required denominator bound. The loss is $1 + \varepsilon$, so taking $\varepsilon \downarrow 0$ gives an isometry.

4 The denominator transfer

We use the standard principle of local reflexivity in the following form: if $M \subset F''$ and $N \subset F'$ are finite-dimensional and $\varepsilon > 0$, there is a linear $T : M \rightarrow F$ such that

$$\|T\| \leq 1 + \varepsilon, \quad f'(Ty'') = y''(f') \quad (y'' \in M, f' \in N). \quad (4)$$

See, for example, Oja and Pöldvere [3].

Lemma 1 (Finite denominator transfer). *Let $y''_1, \dots, y''_m \in F''$, let $M = \text{span}\{y''_1, \dots, y''_m\}$, and let $T : M \rightarrow F$ satisfy $\|T\| \leq 1 + \varepsilon$. For arbitrary $x_{ij} \in E_i$,*

$$\begin{aligned} & \sup_{\substack{x'_i \in B_{E'_i} \\ f' \in B_{F'}}} \left(\sum_{j=1}^m |x'_1(x_{1j}) \cdots x'_n(x_{nj}) f'(Ty''_j)|^q \right)^{1/q} \\ & \leq (1 + \varepsilon) \sup_{\substack{x'_i \in B_{E'_i} \\ g' \in B_{F'}}} \left(\sum_{j=1}^m |x'_1(x_{1j}) \cdots x'_n(x_{nj}) y''_j(g')|^q \right)^{1/q}. \end{aligned}$$

Proof. Fix $f' \in B_{F'}$. The functional $f' \circ T$ on M has norm at most $1 + \varepsilon$. By Hahn–Banach it extends to a $\Phi \in F'''$ with $\|\Phi\| \leq 1 + \varepsilon$. Goldstine’s theorem applied to F' says that $J_{F'}(B_{F'})$ is weak-star dense in $B_{F''}$. Consequently there is a net $g'_\alpha \in (1 + \varepsilon)B_{F'}$ such that

$$y''_j(g'_\alpha) \longrightarrow \Phi(y''_j) = f'(Ty''_j) \quad (j = 1, \dots, m).$$

For every fixed tuple $x'_i \in B_{E'_i}$, continuity of a finite ℓ_q norm and homogeneity now give

$$\begin{aligned} & \left(\sum_{j=1}^m |x'_1(x_{1j}) \cdots x'_n(x_{nj}) f'(Ty''_j)|^q \right)^{1/q} \\ & \leq (1 + \varepsilon) \sup_{g' \in B_{F'}} \left(\sum_{j=1}^m |x'_1(x_{1j}) \cdots x'_n(x_{nj}) y''_j(g')|^q \right)^{1/q}. \end{aligned}$$

Taking the remaining suprema proves the lemma. $\square$

5 Main theorem

Theorem 2. *For all real or complex Banach spaces $E_1, \dots, E_n, F$ and all $1 \leq q \leq p < \infty$,*

$$\mathcal{L}_{\tau(p;q)}(E_1, \dots, E_n; F') = \mathcal{L}_{q\tau(p;q)}(E_1, \dots, E_n; F')$$

isometrically. In particular, for all Banach spaces E, F ,

$$\boxed{\mathcal{L}_{\tau(p)}(E; F') = \mathcal{L}_{q\tau(p)}(E; F') \quad \text{isometrically.}}$$

Proof. Only the reverse of (3) needs proof. Let S be quasi- $\tau(p; q)$ -summing and fix finite families $x_{ij} \in E_i$ and $y''_j \in F''$. Put

$$M = \text{span}\{y''_1, \dots, y''_m\}, \quad N = \text{span}\{S(x_{1j}, \dots, x_{nj}) : 1 \leq j \leq m\} \subset F'.$$

For $\varepsilon > 0$, choose $T : M \rightarrow F$ as in (4), and set $z_j = Ty''_j$. Preservation of the pairings gives

$$S(x_{1j}, \dots, x_{nj})(z_j) = y''_j(S(x_{1j}, \dots, x_{nj})) \quad (j = 1, \dots, m).$$

Apply the quasi inequality (1) to the families x_{ij} and z_j . Lemma 1 bounds its right-hand side by $1 + \varepsilon$ times the right-hand supremum in (2). Hence S satisfies (2) with constant

$$(1 + \varepsilon) \|S\|_{q\tau(p;q)}.$$

Since $\varepsilon > 0$ is arbitrary, $\|S\|_{\tau(p;q)} \leq \|S\|_{q\tau(p;q)}$. Together with (3), this proves equality of the spaces and norms. $\square$

6 Consequences and verification notes

The source paper needed F reflexive to identify the two classes. Theorem 2 removes that assumption completely. Substituting the theorem into Botelho–Mujica’s Borel-transform result therefore replaces quasi- $\tau(p)$ by ordinary $\tau(p)$ for arbitrary F , under exactly the same bounded-approximation hypotheses on the domain duals as in their Theorem 3.4.

The proof is valid for $p = q = 1$, for all finite p, q in the stated range, and over both scalar fields. No sequential compactness is used: Goldstine provides a net, and only its values on finitely many vectors enter. If the ordinary denominator is zero, Lemma 1 forces the transferred quasi denominator to be zero, so that edge case is also covered. There is no computational dependency.

Human review should concentrate on two points: (i) the specialization of Mujica’s definition to target F' , which makes the ordinary test vectors members of F'' ; and (ii) Lemma 1, especially the use of Goldstine for the canonical embedding $F' \hookrightarrow F'''$. These are the only type-sensitive steps.

7 Bounded novelty search and limitations

The search through 13 August 2026 included the run’s result and attempt indexes; exact-title and arXiv-id searches; the phrases “quasi- $\tau(p)$ -summing”, “ $\tau(p)$ -summing local reflexivity”, and “ideal extensions quasi- $\tau(p)$ ”; Botelho and Mujica’s publications; and all five works indexed by OpenAlex as citing the 2017 article. The authors’ 2019 follow-up [4] restates the same isometric question and explicitly treats its answer as unknown. No searched later source states the theorem above or a local-reflexivity proof of it.

Accordingly this is classified as a candidate new full solution, not a known literature answer. The search is bounded rather than exhaustive, so novelty confidence is moderate and specialist bibliographic review remains necessary. The mathematical argument is self-contained modulo the standard local reflexivity, Hahn–Banach, and Goldstine theorems.

References

- [1] G. Botelho and X. Mujica, *Spaces of $\sigma(p)$ -nuclear linear and multilinear operators on Banach spaces and their duals*, Linear Algebra Appl. 519 (2017), 219–237; arXiv:1608.06926. DOI: [10.1016/j.laa.2017.01.002](https://doi.org/10.1016/j.laa.2017.01.002).
- [2] X. Mujica, *$\tau(p; q)$ -summing mappings and the domination theorem*, Portugaliae Math. 65 (2008), 211–226. DOI: [10.4171/PM/1806](https://doi.org/10.4171/PM/1806).
- [3] E. Oja and M. Põldvere, *Principle of local reflexivity revisited*, Proc. Amer. Math. Soc. 135 (2007), 1081–1088. DOI: [10.1090/S0002-9939-06-08612-6](https://doi.org/10.1090/S0002-9939-06-08612-6).
- [4] G. Botelho and X. Mujica, *Ideal extensions of classes of linear operators*, Studia Math. 247 (2019), 285–297. DOI: [10.4064/sm171030-15-3](https://doi.org/10.4064/sm171030-15-3).